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# ME 1060: DYNAMICS

## Lecture 5: Kinetics of a Particle:

## Conservation of energy

# Overview

- Conservative Forces
- Potential Energy
- Gravitational Potential Energy
- Elastic Potential Energy
- Conservation of Energy

# Conservative Forces

If the work of a force is independent of the path and depends only on the force's initial and final positions on the path, then we can classify this force as a conservative force.

Examples: The weight of a particle, and the force developed by a spring.

In contrast, Friction force is a non conservative force as it depends on the ***path.*** The work is dissipated from the body in the form of heat.

# Energy

Energy is defined as the capacity for doing work.

## Kinetic Energy

The kinetic energy is equal to the work that must be done on the particle to bring it from a state of rest to a speed v. Thus, the kinetic energy is a measure of the particle's capacity to do work, which is associated with the motion of the particle.

## Potential Energy

When energy comes from the position of the particle, measured from a fixed datum or reference plane, it is called potential energy.

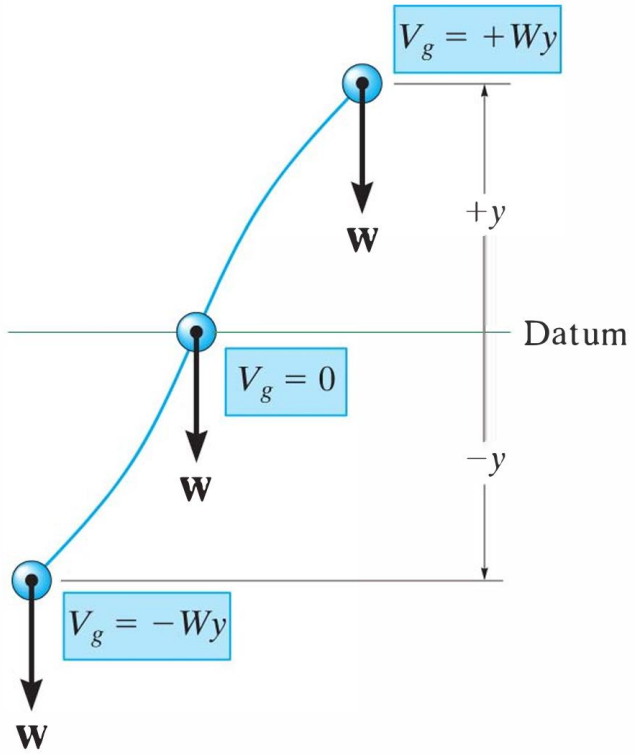
Thus, potential energy is a measure of the amount of work a conservative force will do when it moves from a given position to the datum.

# Gravitational Potential Energy

- If a particle is located a distance y above an arbitrarily selected datum, particle's weight W has positive gravitational potential energy.
- If the particle is located a distance y below the datum, Vg is negative.
- At the datum Vg = 0

In general, if y is positive upward, the gravitational potential energy of the particle of weight W is;

$$V_g = Wy$$



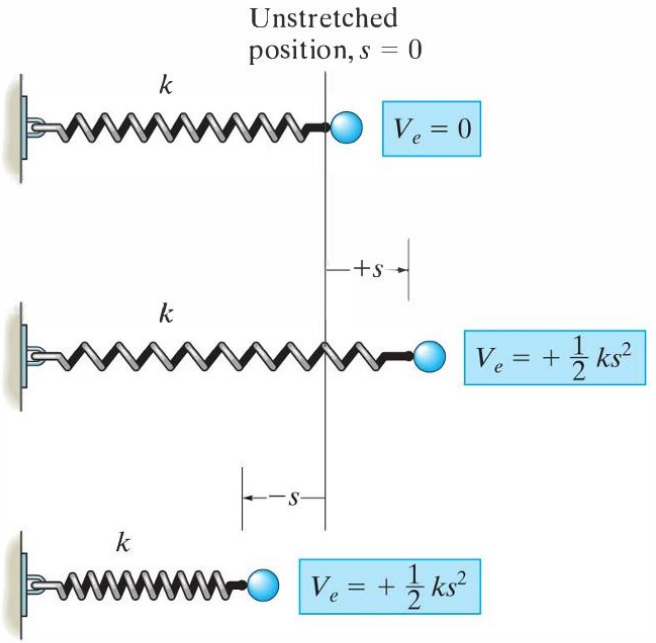
# Elastic Potential Energy

- When an elastic spring is elongated or compressed a distance s from its unstretched position, elastic potential energy Ve can be stored in the spring.

This energy is;

$$V_e = \frac{1}{2}ks^2$$

Here v;, is always positive since, in the deformed position, the force of the spring has the capacity or "potential" for always doing positive work on the particle when the spring is returned to its unstretched position




Elastic potential energy

# Potential Function

In the general case, if a particle is subjected to both gravitational and elastic forces, the particle's potential energy can be expressed as a potential function, which is the algebraic sum

$$V = V_e + V_g$$

The work done by a conservative force in moving the particle from one point to another point is measured by the difference of this function, i.e.,

$$U_{1-2} = V_1 - V_2$$

# Conservation of Energy

When a particle is acted upon by a system of both conservative and nonconservative forces, the portion of the work done by the conservative forces can be written in terms of the difference in their potential energies

$$(\sum U_{1-2})_{cons} = V_1 - V_2$$

As a result, the principle of work and energy can be written as

$$T_1 + V_1 + (\sum U_{1-2})_{noncons} = T_2 + V_2$$

Here $(\sum U_{1-2})_{noncons}$ represents the work of the nonconservative forces acting on the particle.

If **only conservative forces do work**, then

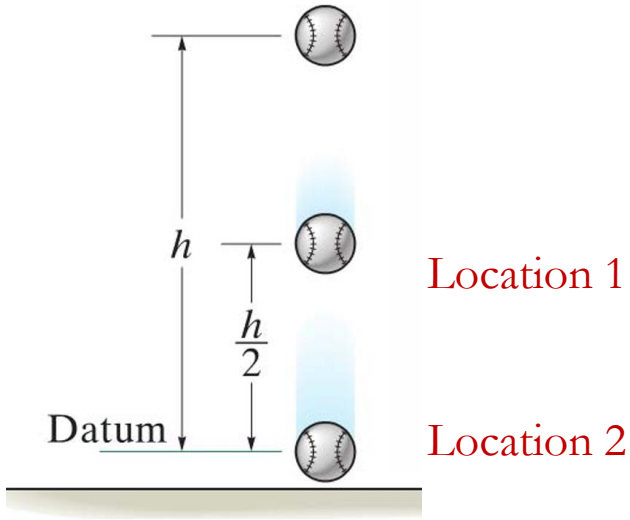
$$T_1 + V_1 = T_2 + V_2$$

During the motion the sum of the particle's kinetic and potential energies remains constant.

This equation is referred to as the conservation of mechanical energy or simply the conservation of energy.

# Conservation of Energy

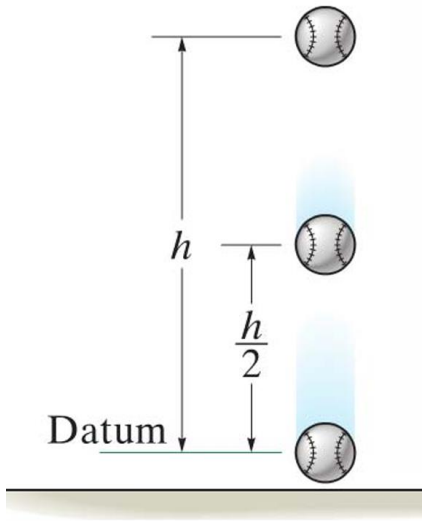
If a ball of weight W is dropped from a height h above the ground (datum), find the velocity of the ball at each location, using energy balance.




when the ball comes in contact with the ground, it deforms somewhat, and provided the ground is hard enough, the ball will rebound off the surface, reaching a new height $h'$.

# Conservation of Energy

If a ball of weight W is dropped from a height h above the ground (datum), find the velocity of the ball at each location, using energy balance.




Location 1

Location 2

**Location 1:**

$$E = T_1 + V_1 = 0 + Wh$$

$$\frac{1}{2}\frac{W}{g}v^2 + \frac{Wh}{2} = Wh$$

$$v = \sqrt{gh}$$

Using Kinematics:

When the ball is a distance h/2, its speed is;

$$v^2 = v_0^2 + 2as = 0 + 2gh$$

$$v = \sqrt{gh}$$

$$v_1 = \sqrt{gh}$$

$$v_2 = \sqrt{2gh}$$

when the ball comes in contact with the ground, it deforms somewhat, and provided the ground is hard enough, the ball will rebound off the surface, reaching a new height $h'$.

# Conservation of Energy for a System of Particles

If a system of particles is subjected **only to conservative forces** then,

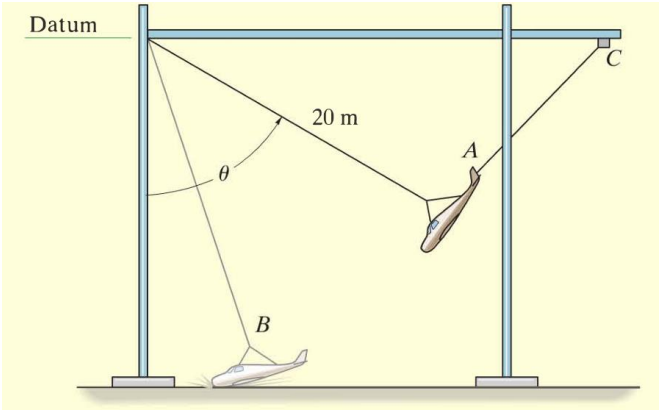
$$\sum T_1 + \sum V_1 = \sum T_2 + \sum V_2$$

Here, the sum of the system's initial kinetic and potential energies is equal to the sum of the system's final kinetic and potential energies.

# Example 1

The gantry structure in the photo is used to test the response of an airplane during a crash. As shown in figure, the plane, having a mass of 8 Mg, is hoisted back until $\theta = 60°$, and then the pull-back cable AC is released when the plane is at rest. Determine the speed of the plane just before it crashes into the ground, $\theta = 15°$. Also, what is the maximum tension developed in the supporting cable during the motion? Neglect the size of the airplane and the effect of lift caused by the wings during the motion.





# Example 1

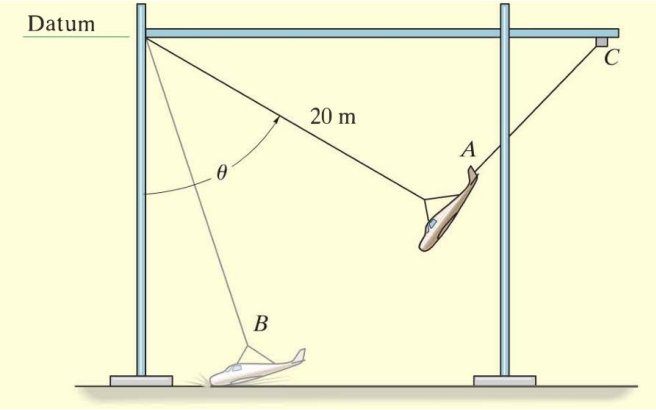
For convenience, the datum has been established at the top of the gantry.

Conservation of Energy:

$$T_A + V_A = T_B + V_B$$

$$0 \; - 8000\,(9.81)(20\cos 60) = \frac{1}{2}(8000)v_B^2 \; - 8000(9.81)(20\cos 15^o)$$

$$v_B = 13.52\; ms^{-1}$$




Equation of Motion:

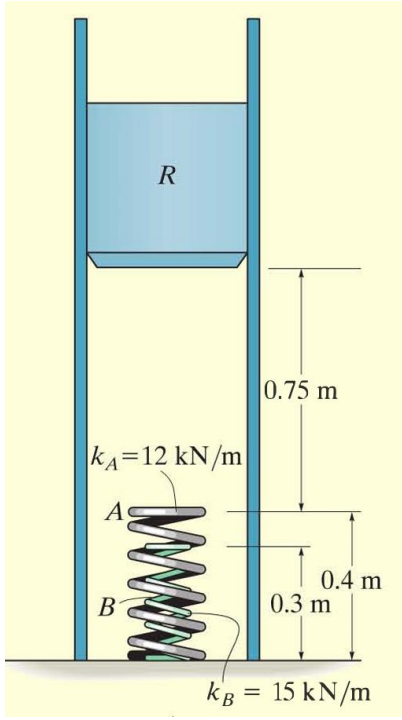
$$F = ma_n$$

$$T \; - 8000(9.81)\cos 15^o = (8000)\left(\frac{13.52}{20}\right)$$

$$T = 149\; kN$$

# Example 2

The ram R shown in figure has a mass of 100 kg and is released from rest 0.75 m from the top of a spring, A, that has a stiffness $k_A = 12\,kN/m$. If a second spring B, having a stiffness $k_B = 15\,kN/m$, is "nested" in A, determine the maximum displacement of A needed to stop the downward motion of the ram.
The unstretched length of each spring is indicated in the figure. Neglect the mass of the springs.




## Potential Energy

We will assume that the ram compresses both springs at the instant it comes to rest. The datum is located through the center of gravity of the ram at its initial position. When the kinetic energy is reduced to zero ($v_2 = 0$), A is compressed a distance $s_A$ and $B$ compresses $s_B = s_A - 0.1\,m$.

# Example 2

## Conservation of Energy

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 0 = 0 + \{\frac{1}{2}k_A s_A^2 + \frac{1}{2}k_B(s_A - 0.1)^2 - Wh\}$$

$$0 + 0 = 0 + \left\{\frac{1}{2}(1200)s_A^2 + \frac{1}{2}(15000)(s_A - 0.1)^2 - 981(0.75 + s_A)\right\}$$

Rearranging the terms;

$$13500\, s_A^2 - 2481\, s_A - 660.75 = 0$$

Solving for the positive roots:

$$s_A = 0.331\, m$$

$$s_B = 0.331\, m - 0.1\, m = 0.231\, m$$

